Building GPS orbits from station data

Final report: a self-written GPS orbit estimator, its validation against NGA PE, and its limits

customKF | Final report | 16 October 2026 UTC | results as recorded by 8 October 2026

Summary. A retrospective proof of concept: our own Python GPS orbit estimator, run on one Windows PC.

- **Inputs:** about 25 public IGS stations, the broadcast navigation message, published models and calibrations.
- **Output:** 24-hour orbits for the healthy GPS satellites with adequate navigation support.
- **The bar:** be closer to IGS final than the NGA precise ephemeris (NGA PE) is, on the same satellite-epochs. IGS final is a reference, not truth.
- **Frozen default** (section 4): 25 stations, 24-hour arcs, ECOM5 solar pressure, Earth radiation and antenna thrust, IERS sub-daily EOP, elevation weighting.

Result. On five validation dates drawn by a pre-registered rule, the frozen default configuration was closer to IGS final than NGA PE on every date, in orbit-only URE (the primary metric) and in 3D RMS.

Five validation dates	Mean URE (cm)	Mean 3D RMS (cm)	Mean ratio to NGA PE: URE	Mean ratio to NGA PE: 3D	Dates below NGA PE (URE, 3D)
Default	3.06	8.97	0.766	0.855	5/5, 5/5
Default + gradients (final latency)	3.00	8.76	0.752	0.835	5/5, 5/5
+ gradients, 50 stations instead of 25	2.55	7.16	0.638	0.683	5/5, 5/5
+ gradients, 17 stations near NGA sites	4.83	15.14	1.215	1.444	1/5, 0/5
NGA PE, same satellite-epochs	4.01	10.50	1	1	-

- **Why the 3D margin is smaller.** URE is mostly radial error; 3D RMS is mostly along-track. Averaged over the validation dates, our radial error is 25% below NGA PE's and our cross-track error 26% below, but our along-track error is 3% above it (section 11).
- **How often.** Our constellation URE is below NGA PE's at 92.0% of 15-minute epochs over the validation dates and Jun 27 (section 15).
- **Network size.** 50 stations cut URE 15.0% and 3D RMS 18.4% against 25; 17 stations did not beat NGA PE (section 19).

> **Scope.** Retrospective results on summer 2026 dates. Every input was downloaded after the fact and no rapid-deadline replay was done (section 16). The gradient and station rows are final latency and reuse the validation dates, so they are contrasts, not further validations (section 21).

From the first station-fitted orbits (158 m 3D RMS: 8 stations, 3-hour arc) to the validated default (8.97 cm 3D, 3.06 cm URE), dates, arcs and settings changed, so this shows the scale of the progress, not one controlled contrast.

Contents: goal, metrics and system 1-5; development 6-10; validation against NGA PE 11-15; latency and independence 16-17; station tests 18-19; engineering 20; scope, conclusions and future work 21-22; evidence, references and document control 23-24.

1. The goal, the bar and the metrics

The product

GPS-only orbits estimated from station observations (OBS) and broadcast navigation (NAV), with inputs that could plausibly be available within about a day (a "rapid" product). Our own estimator does the work; no GNSS processing package is used. It does rely on external coordinates, calibrations, Earth orientation and model data (section 21). IGS describes its rapid orbits as available about 17 hours after the day ends. R1, IGS products.

The bar: NGA PE against IGS final

NGA's precise ephemeris is an independent operational product. Our target is to be closer to IGS final than NGA PE is, on exactly the same satellite-epochs. IGS final is a combined, high-quality reference, but it is not truth; "closer to IGS final" means closer to that reference.

Metrics

Every 24-hour comparison with NGA PE (sections 8-19) is scored this way. The development milestones of sections 6-7 are mostly 3- and 6-hour fits scored in 3D RMS against IGS rapid; each states its arc, metric and reference.

- **Comparison.** Raw positions at IGS final's 15-minute epochs, 00:00-23:45 GPST. No alignment, no mean removal, no satellite dropped for scoring badly. Each solution is rotated to Earth-fixed axes with the Earth-orientation model it was fitted with. Only satellite-epochs present in every compared arm and in NGA PE are scored.
- **Orbit-only URE (primary)** is sqrt(mean(0.98^2 R^2 + (A^2 + C^2)/49)), with R, A, C the radial, along-track and cross-track errors in the reference orbit's axes. Clocks are not included: ours are nuisance parameters.
- **3D RMS (secondary)** is the RMS of the full position error on the same satellite-epochs.
- **Why they differ.** A user's range error comes mostly from the radial error, so URE weights it almost fully and the other two by 1/49. 3D RMS weights all three equally and is dominated by along-track, our weakest component. Our gains are therefore larger in URE than in 3D, and an orbit can beat NGA PE in URE but not in 3D (Aug 2, section 8).
- **Ratio to NGA PE:** ours divided by NGA PE's, per date, on the same satellite-epochs. Below 1 is closer to IGS final.
- **Epoch share:** the fraction of 15-minute epochs at which our constellation URE (the RMS over all satellites at that epoch) is below NGA PE's. The **satellite-epoch share** counts each satellite at each epoch separately.
- **Averages.** A mean over dates is the equal-weight mean of the daily values. A mean ratio or mean change is the mean of the daily ratios or changes, not one computed from the mean errors. Shares are pooled counts.
- **Reporting.** Results give URE and 3D RMS together; where one is quoted alone it is named. Errors are in cm to two decimals (metre-scale development results in m; tiny diagnostics in mm or micrometres), ratios to three decimals, changes in percent to one decimal (two where small or near a decision threshold).

Dates

- **Development days:** 2-5 and 7 August 2026. Used to build and select the model.
- **Validation dates:** 6, 20 and 23 June, 6 and 23 July 2026. Drawn by a pre-registered random rule after the default was frozen (section 11); reused afterwards for the gradient and station tests.
- **Jun 27, 2026:** one further date, first used for the gradient model (section 12).

Production scores come from `Opus_Fork/evaluate.py`. A separately written scorer, `validation/oracle/`, reproduces them within 0.1 mm (section 20).

2. How the system works

The pipeline

1. **Fetch.** Daily 30-second RINEX 3 observation files for each station, and the broadcast navigation for the day and the previous day, from the IGS data centre at BKG. Static models and calibrations are fetched once.
2. **Prepare.**
   - Choose a dual-frequency signal pair; track phase arcs on the native 30 s data; form ionosphere-free code and phase every 600 s above 10 degrees elevation.
   - Apply station coordinates, tides, ocean loading and antenna calibrations.
   - Select healthy broadcast records; compute each satellite's starting state from the broadcast message.
   - Freeze everything, hashed, into one fixture.
3. **Fit.** A batch least-squares estimator, iterated to convergence, with each iteration as a separate guarded worker:
   - integrate every satellite's orbit and its sensitivities over 24 hours;
   - model every observation;
   - fold the observations, chunk by chunk, into a square-root information system;
   - solve and line-search.
4. **Score** against IGS final, with NGA PE beside it.
5. **Export** an SP3 file through a verified temporary copy; an existing product is never overwritten.

Why a batch square-root estimator

A batch fit uses all the data over the arc at once. It exposes weak directions, prior dependence and parameter coupling, which matter with a small network. Square-root (QR) processing avoids squaring the condition number, as normal equations would. Receiver and satellite clocks are eliminated epoch by epoch, so the retained system stays small. For large networks an option folds each phase ambiguity out once its arc has ended (section 18).

A Kalman filter or square-root information filter can represent the same information sequentially. We kept the batch form for its diagnostics; this choice was not tested against a filter. R2, GROOPS cookbook: GNSS network processing.

What runs where

- **Code.** `customkf/` holds the core physics, time and frames. `claude/roadmap/` holds the 24-hour streaming estimator. `Opus_Fork/` holds the production pipeline (fetch, store, preparation, fit controller, models, evaluator, exporter). `validation/oracle/` holds the independent scorer.
- **How jobs run.** Numerical campaign work runs under resource guards on one Windows PC, as resumable chains, because the PC is switched off overnight (section 20).
- **Further reading.** Setup: `SETUP.md`. The technical specification: `docs/REPRODUCING_WITH_AN_LLM.md`.

3. Models used: forces

Force	Model in the default
Gravity	EGM2008 to degree and order 12; solid-Earth tides from IERS 2010 (anelastic Love numbers, degrees 2-3) [R3, R9]
Sun and Moon	JPL DE440s positions, direct minus indirect attraction [R8]
Relativity	Schwarzschild term
Solar radiation	ECOM5: a nominal Sun-direction acceleration plus five estimated terms per satellite per arc (section 5). Earth shadow as a penumbra cone [R4]
Earth radiation	Uniform Earth: 238 W/m2 infrared and 0.3 albedo with a Lambert phase law. Absorption only, on published box-wing areas
Antenna thrust	Transmit power / (mass x c), radially outward, from the IGS satellite metadata
GPS-IIIA mass	2,600 kg, the published effective value [R7] (metadata: 2,161 kg)
Attitude	Nominal yaw steering; no eclipse yaw manoeuvres

Integration. DOP853 (SciPy) with the state transition and force sensitivities integrated alongside: maximum step 60 s, relative tolerance 1e-12. Synthetic two-body tests agree to 0.5 micrometres over 24 hours, and tested derivatives agree with finite differences to about 13 micrometres. These bound the tested cases only; the full adopted model has not been reintegration-checked at its real boundaries (section 22).

Earth orientation. IERS daily EOP, plus the IERS 2010 sub-daily ocean-tide model (ORTHO_EOP) for polar motion and UT1 [R3]. It is applied in the orbit dynamics, the station geometry, the evaluator and the exporter alike. Libration terms are omitted.

What is fixed and what is estimated

Fixed:

- the Earth-radiation and thrust forces, computed from published geometry and metadata, never fitted;
- the nominal solar-pressure acceleration.

Estimated per satellite:

- the six initial position and velocity corrections to the broadcast-derived start;
- the five solar-pressure terms.

The estimated terms have no priors. The position and velocity priors (1 km, 1 m/s) are deliberately loose.

Omitted:

- frequency-dependent and ocean tides in the gravity field;
- eclipse attitude;
- reflection in Earth radiation;
- empirical along-track or cross-track accelerations.

These omissions are candidates for the remaining along-track error, not identified causes.

4. Observations and estimation

Element	Model
Observables	Ionosphere-free code (C1W/C2W) and carrier phase (L1/L2), every 600 s, 10-degree cutoff
Geometry	Light time iterated in the inertial frame; Shapiro delay; relativistic clock correction
Antennas	IGS20 ANTEX phase-centre offsets and variations for satellites and receivers; phase wind-up
Troposphere	Saastamoinen hydrostatic delay from standard pressure; Niell mapping; wet delay estimated per station every 3 h
Station motion	IGS20 coordinates and velocities; solid-Earth tide; pole tide; FES2014b ocean loading [R10]
Weights	Code 0.9 m / sin(e). Phase 9 mm / sin(e)^1.3 (elevation weighting)
Clocks	Receiver and satellite clocks per epoch, eliminated before the solve; BRUX is the clock datum
Ambiguities	Float, one per phase arc; no integer resolution
New phase arc	New satellite; a gap over 60 s in the native data; loss of lock or half-cycle flags; receiver event or slip records; geometry-free or Melbourne-Wubbena jumps

Iterating to a solution

Starting point. Each fit starts from the broadcast orbit at midnight. The broadcast position refers to the antenna phase centre, so it is converted to the centre of mass.

Each iteration:

1. Integrate.
2. Linearize the observations.
3. Accumulate them in chunks of 12 epochs, each a separate guarded worker.
4. Solve for a Gauss-Newton step.
5. Try step fractions 1, 1/2, 1/4 and 1/8 with full reintegration, and keep the first that lowers the cost.

The fit stops when the step moves the orbit by less than 1 mm and the predicted cost reduction is negligible. Typical fits converge in three or four iterations, about an hour of guarded computing for 25 stations. That is a local stationarity rule; it certifies neither the physics nor a unique optimum.

The frozen default configuration

25 stations; 24-hour arc; previous-day navigation, with satellites kept whenever navigation allows; ECOM5 solar pressure; Earth radiation and antenna thrust; 2,600 kg GPS-IIIA mass; IERS sub-daily EOP; elevation weighting. In the code: `erp+thrust+iiia2600+sdeop+elw13` with `unit-ecom5.json`. Every element was adopted under its own written rule; section 9 shows how much each one helped.

Optional model (final latency only): add TU Wien GRAD operational tropospheric gradients (section 12).

5. Once- and twice-per-rev terms

The solar-radiation model is ECOM (the Empirical CODE Orbit Model) [R4]. In our code **D** is the unit vector from the satellite to the Sun, **Y** = r x D normalised (r the unit position vector), and **B** = D x Y. The angle **u** is measured in the orbit plane from the ascending node in ECOM5, and from the Sun's projection on the orbit plane in the seven-term variant. These are separate from the antenna attitude conventions.

Term	Acceleration	Period	Status
D0, Y0, B0	Constant along D, Y, B	-	Estimated in every fit
Bc1, Bs1	cos u and sin u, along B	**once per revolution**	**Estimated in every 24-hour fit, for every satellite, with no prior**
Dc2, Ds2 (ECOM-7)	cos 2u and sin 2u, along D	**twice per revolution**	Tested twice, rejected both times
Along-/cross-track 1/rev	Empirical once-per-revolution pulses	once per revolution	Never used

Terms estimated on each arc

- **All five ECOM5 terms**, including the two once-per-revolution terms. They are estimated anew for every satellite on every 24-hour arc.
- **Shorter arcs used less.** The 6-hour development fits estimated only D0, or nothing. On those arcs the once-per-revolution terms are poorly separable from the starting state.
- **No twice-per-revolution term, and no along-track or cross-track empirical acceleration, is in any adopted configuration.**

What the twice-per-revolution tests showed

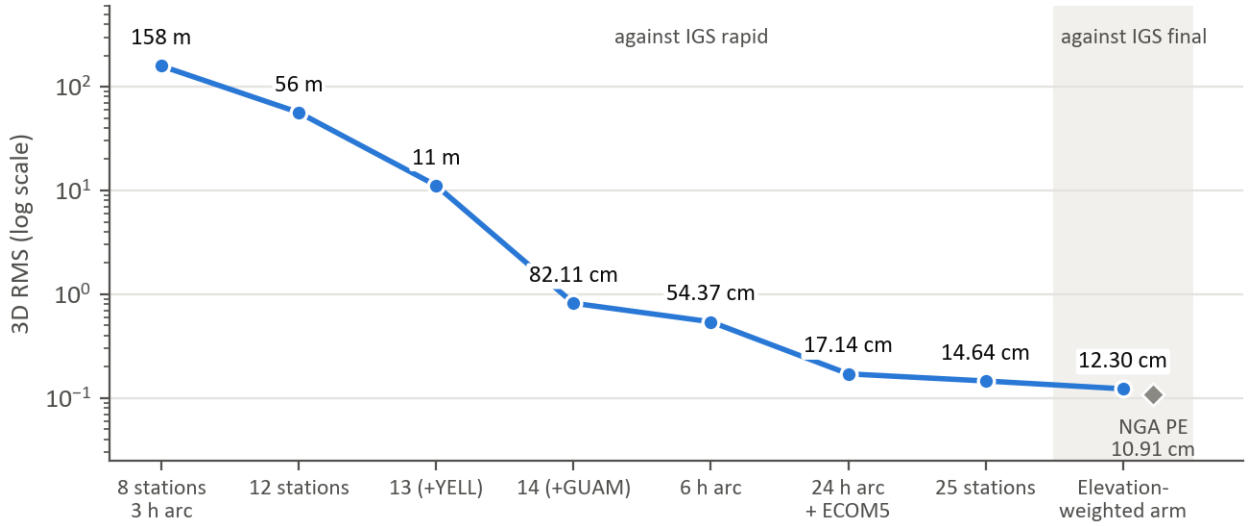
Our orbits had a 12-hour radial "swing" against IGS. A twice-per-revolution D-term was one candidate; a 12-hour pattern does not identify its cause uniquely.

- **Free ECOM-7:** the swing fell 30-37%, but URE worsened 29.5% (Aug 2) and 16.9% (Aug 3). Rejected.
- **ECOM-7 with 2 nm/s2 priors:** URE on Aug 2-5 improved 2.9-6.1%, but Aug 7 worsened 4.5% and the hold-out rule failed. The prior width came from fits to IGS orbits, so it was also reference-informed. Rejected.
- **Adding the IERS sub-daily Earth rotation** cut URE by 13.7% and 3D RMS by 10.1% on average over the development days, and reduced the 12-hour amplitude on Aug 2-5. On Aug 7 the amplitude rose from 0.47 to 1.07 cm. It was a major contributor on those dates, not a proven sole cause (section 9).

What remains: a residual 12-hour scale "breathing" of 0.16-0.37 ppb, which NGA PE also shows against IGS. A box-wing solar model was implemented but never validated in the station fit.

Evidence: `ECOM7_RESULTS.md`, `ECOM7P_RESULTS.md`, `ECOM7P_HOLDOUT_RESULTS.md`, `SDEOP_RESULTS.md`, in `Opus_Fork/results/`.

6. Development: from 158 m to 15 cm



Development milestones on 2 August 2026, 3D RMS on a log scale. Up to the 25-station fit the reference is IGS rapid (the metric and reference then used to steer development); the last point, the Aug 2 elevation-weighted arm (every adopted model change, but IIIA mass 2,161 kg that day), is against IGS final, beside NGA PE on the same 31 satellites. Arc length, sampling and settings changed along the way: the points show the scale of the progress, not exact step effects.

Step (Aug 2)	3D RMS (IGS rapid)	What it taught us
First station-fitted orbits, 8 stations, 3 h arc	158 m	A 7 mm phase residual came with a 158 m orbit. Weakly viewed satellites absorb errors
Starting-state fix (antenna phase centre to centre of mass)	157 m	A real bug: start error 27 m to 1.2 m, but not the cause of the poor fit
3 h arcs: 12, then 13 (+YELL), then 14 stations (+GUAM)	56 m, 11 m, 82.11 cm	**Geometry was the largest single gain**: which satellites are seen together, from where
6-hour arc instead of 3	54.37 cm	Longer tracking separates orbit from nuisance parameters
24-hour arc + ECOM5, 14 stations	17.14 cm (URE 6.09 cm)	Long arcs let all five solar-pressure terms be estimated
25 stations (greedy geometry screen)	14.64 cm (URE 4.64 cm)	The network every later result uses

URE is given from the 24-hour fits on, where it was computed on the convention of section 1.

The 25-station network was chosen by a greedy information screen that added 11 stations to the original 14:

- the original 14: BRUX, HOB2, KOUR, ABMF, KOUG, DJIG, NNOR, MAS1, DAV1, FALK, FAA1, GLPS, YELL, GUAM;
- added: SGOC, KOKB, SUTH, KIT3, KIRI, NKLG, MDO1, EUR2, VACS, PIMO, CHPI.

Against IGS final at that point (the 27 satellites both products had): URE 4.63 cm against NGA PE's 4.43 cm, within 5%; 3D RMS 14.67 cm against 10.84 cm. The gap was along-track (12.04 against 6.49 cm), which URE weights lightly.

Evidence: `STAGE5_RESULTS.md`, `STAGE5B_RESULTS.md`, `YELL_GUAM_RESULTS.md`, `ARC_LENGTH_RESULTS.md`, `claude/roadmap/STEP3_RESULTS.md`, `claude/roadmap/STATIONS_RESULTS.md`, `Opus_Fork/results/BASELINE.md`.

7. Development: lessons and failures

Things that did not work, did not replicate or barely helped

Experiment (3D RMS against IGS rapid)	Outcome
One estimated solar-pressure constant (D0) per satellite; 6 h arcs	Aug 2: 54.37 to 44.18 cm (-19%). **Aug 3: 62.08 to 71.35 cm (+15%).** Did not replicate; one satellite (G27) dominated
Removing the late GUAM gap on Aug 2 (control); 6 h	Did not reproduce the Aug 3 failure; G27 was not explained by that gap
Hourly wet-delay nodes; 3 h, 14 stations	0.3% gain with six large regressions; not adopted
A 15th station (PIMO); 3 h	0.3% gain; later part of the 25
Uniformly stronger phase weights; 3 h, 14 stations	96.21 cm against 82.11 cm; worse
Block-mean D0 from IGS rapid; 6 h arcs	Large gains, but **reference-informed**; superseded by 24 h + ECOM5

What we learned

- **Small residuals do not mean a good orbit.** With weak geometry, clocks, ambiguities and weak orbit directions absorb errors.
- **One good day is not a result.** The D0 failure on Aug 3 is why every later change needed a second day, and finally untouched dates.
- **Checks bound errors; they do not identify causes.** Independent checks bounded the tested numerical errors. Fitting ECOM5 directly to IGS positions over 6 hours left a withheld position RMS of 0.84 cm (31 satellites), so the force basis could approximate those arcs closely, but that did not identify the causes of the station-fit error.
- **Later defects still happened.** An antenna-derivative defect, found during validation, needed repair and separate validation (section 11).

The comparison rule that came out of it

- Change one thing at a time against a matched same-day arm.
- Write the prediction and the pass/fail rule before the fit.
- Score all satellites, with no exclusions.

From the 24-hour comparisons on, the rule was enforced: protocols are pinned into each campaign's records before it runs.

Evidence: `D0_POD_RESULTS.md`, `D0_DAY215_RESULTS.md`, `GUAM_GAP_RESULTS.md`, `WET_DELAY_EXPERIMENT.md`, `FORCE_PARAMETER_RESULTS.md`, `claude/roadmap/BLOCK_D0_RESULTS.md`.

8. Development: past NGA PE

The production code line (`Opus_Fork/`) ran 24-hour fits against IGS final with NGA PE on the same satellite-epochs. Each step below was one matched same-day change on the development days. Values are URE / 3D RMS in cm.

Step (adopted)	Aug 2	Aug 3	Aug 4	Aug 5	Aug 7
Baseline	4.43 / 14.11	4.26 / 11.05	4.33 / 11.53	4.23 / 11.52	3.56 / 11.50
+ Earth radiation, antenna thrust (and IIIA mass)	4.17 / 14.04	3.88 / 10.94	4.04 / 11.43	3.84 / 11.43	3.40 / 11.49
+ IERS 2010 sub-daily EOP	3.40 / 12.92	3.16 / 9.67	3.36 / 10.41	3.25 / 9.69	3.41 / 10.71
+ elevation weighting	3.30 / 12.30	3.04 / 9.28	3.22 / 9.69	3.20 / 9.19	3.27 / 10.12
NGA PE	4.43 / 10.91	4.29 / 10.78	4.36 / 10.93	4.83 / 11.76	4.19 / 10.56

A lineage, not one fixed configuration. The last row of arms is called the elevation-weighted arm, not the default: the IIIA mass was 2,161 kg on Aug 2 and 7 and 2,600 kg on Aug 3-5. Aug 4 and 5 had 23 stations (no CHPI, no YELL); the other days 25. Aug 7 used a different preparation (26 satellites).

Before the baseline (Aug 2): the first 25-station 24-hour fit scored 4.63 cm URE on 27 satellites; previous-day and per-epoch navigation eligibility recovered 31 (G13 is always excluded as unhealthy). That changed the population, not the error: the 31-satellite baseline tied NGA PE in URE (4.427 against 4.427 cm).

What each step was:

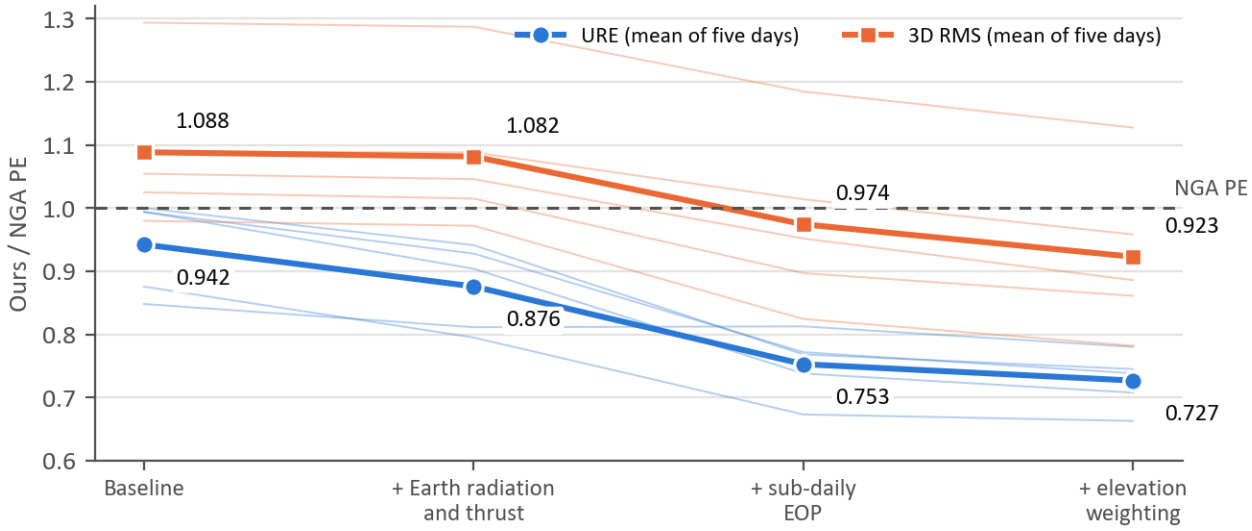
- **Earth radiation and antenna thrust:** fixed physical forces from published geometry and metadata, with nothing fitted. Aug 3-5 also used the published effective IIIA mass, so that step is a composite.
- **Sub-daily EOP:** the IERS 2010 ocean-tide model for polar motion and UT1. It was missing at first, and adding it was the largest single gain (section 5).
- **Elevation weighting:** phase sigma 9 mm/sin(e)^1.3. The exponent came from our own post-fit residuals on Aug 3 (1.27); it was not taken from IGS. Choosing to adopt it used IGS-scored outcomes, a selection effect we disclose.

After the last step, URE is below NGA PE's on all five days, and 3D RMS on four of five. Aug 2 is the exception in 3D (12.30 against 10.91 cm), because of its late-day along-track error.

The limit of these days: they were used to build and select the model, so results on them are contrasts, not validation. A transfer claim needed untouched dates (section 11).

Evidence: `Opus_Fork/results/` (`T2_1_ERP_RESULTS.md`, `IIIA_AUG345_RESULTS.md`, `SDEOP_RESULTS.md`, `ELEVATION_RESULTS.md`), and `docs/consolidation/configurations.md`.

9. How much each improvement helped



Ours divided by NGA PE through the adopted steps on the development days: URE (blue) and 3D RMS (orange). Thin lines are the five days (Aug 2, 3, 4, 5, 7); thick lines their mean. Each step is a matched same-day change; below the dashed line we are closer to IGS final than NGA PE is.

Improvement	URE change	3D RMS change	Days	Status
25 stations instead of 14 (24 h, ECOM5; IGS rapid)	6.09 to 4.64 cm	17.14 to 14.64 cm	Aug 2	adopted
Earth radiation + antenna thrust	-5.9% (Aug 2), -4.3% (Aug 7)	-0.5%, -0.1%	2	adopted
+ published IIIA mass 2,600 kg (composite with the above)	-8.3% mean	-0.9% mean	Aug 3-5	adopted
IERS sub-daily EOP	-13.7% mean (-18.4 to +0.2%)	-10.1% mean (-15.2 to -6.8%)	5	adopted
Elevation weighting (held-out days)	-3.2% mean	-5.6% mean	4	adopted
TU Wien GRAD gradients	-4.2% Aug 3; -1.9% mean on the validation dates; -3.8% Jun 27	-2.7%; -2.4%; -2.9%	7	optional, final latency
50 stations instead of 25 (with gradients)	-15.0% mean	-18.4% mean	5	descriptive test; not adopted

Mean ratio to NGA PE on the development days, URE then 3D: 0.942 and 1.088 at the baseline; 0.876 and 1.082 after Earth radiation and thrust; 0.753 and 0.974 after sub-daily EOP; 0.727 and 0.923 after elevation weighting. The forces mainly reduced the radial error, so they moved URE much more than 3D; along-track improved least.

Evidence: the recorded evaluations `Opus_Fork/results/arm_*.json`; tables and figure data `research/report_rev7/`; `Opus_Fork/experiments/networks/RESULTS.md`.

10. What did not work

Tried	Outcome	Decision
Free ECOM-7 (twice-per-rev D terms; section 5)	URE +29.5% (Aug 2), +16.9% (Aug 3)	Rejected
ECOM-7 with 2 nm/s2 priors	URE -2.9 to -6.1% on Aug 2-5, +4.5% on Aug 7	Rejected (failed hold-out)
15-degree cutoff instead of 10	URE held-out mean +0.06%	Rejected
Per-station empirical reweighting (two variants)	Gains only on the day they were tuned on	Not adopted
GPT3 empirical gradients	URE better on 5 of 5 dates, but mean -0.61% (bar: 1%)	Inconclusive
Hourly wet delay (24 h)	URE -1.4% on one development day (diagnostic)	Not adopted
Desai-Sibois sub-daily EOP	RAC RMS within 3% of ORTHO_EOP in a reference census; 12-hour amplitude mixed	Not used
Box-wing a-priori solar model	Census only	Unvalidated

The estimated D0 on 6-hour arcs, which did not replicate, is in section 7.

Diagnostics with small predicted effects

- **Removing satellite antenna variations** worsened URE by 25%: they matter.
- **Specified atmospheric perturbations** had small predicted URE effects in a local linear transfer model, against a 3.165 cm baseline (Aug 3): independent 10 hPa station pressure 0.035 cm (1.1%); common 10 hPa 0.068 cm (2.2%); S2 climatology 0.040 cm (1.3%); 1 mm mapping error at 10 degrees 0.0007 cm. The method was calibrated on one real refit; these do not rule out all such errors.
- **Station-tide model differences** (solid tide and ocean-loading nodal terms) were scored on the 12-hour scale amplitude rather than URE: at most 1.5% of it on Aug 3.
- **Station gradients:** assuming independent errors of 0.5 mm per north and east component, the same model predicted 0.73 cm URE. That was an assumption, not a measured field, and it led to the gradient tests (section 12).

Why rejected items stay rejected

Each test had a written rule before it ran. A rejected configuration is not retuned on the same days. For example, a "use ECOM-7 only when the swing is present" rule was proposed and refused.

Evidence: `docs/consolidation/configurations.md` sections 3-5; `Opus_Fork/results/*_RESULTS.md`; `Opus_Fork/experiments/atmosphere/RESULTS.md`; `Opus_Fork/results/CORRECTIONS.md`.

11. Validation on untouched dates

Why untouched dates. Every development day had shaped the model choices. So before any download, a protocol froze the pipeline, the success rule and a seeded random stream of dates from 1 June to 27 August. Dates were taken from the stream in order and replaced only for data reasons, such as incomplete navigation files. URE was the registered primary metric; 3D RMS the secondary one.

Date	URE: ours / NGA PE (cm)	URE ratio	3D RMS: ours / NGA PE (cm)	3D ratio	Along-track: ours / NGA PE (cm)
Jun 6	2.91 / 3.80	0.765	9.17 / 10.30	0.890	7.05 / 5.50
Jun 20	3.39 / 3.91	0.868	9.45 / 10.22	0.925	6.16 / 6.68
Jun 23	2.96 / 4.31	0.688	8.42 / 10.63	0.792	5.62 / 6.95
Jul 6	2.91 / 3.80	0.767	9.06 / 10.55	0.859	7.06 / 6.12
Jul 23	3.13 / 4.21	0.743	8.77 / 10.82	0.811	6.36 / 6.09
Mean	**3.06 / 4.01**	**0.766**	**8.97 / 10.50**	**0.855**	**6.45 / 6.27**

Result: below NGA PE in URE and in 3D RMS on all five dates.

- Radial and cross-track errors are below NGA PE's on every date (means 2.87 against 3.83 cm, and 5.50 against 7.47 cm).
- Along-track is worse than NGA PE's on Jun 6, Jul 6 and Jul 23. That is why the 3D margin (mean ratio 0.855) is smaller than the URE margin (0.766).

Qualifications:

- **The repair.** Jun 23 first stopped on a code defect: a numerical derivative crossed the edge of an antenna calibration table. It was fixed and refitted under a documented amendment. The original as-registered result is therefore "not established"; this is the transparently repaired analysis.
- **Model selection.** The configuration was chosen on the development days using IGS-scored outcomes. These dates test transfer, but they are one season and one network.
- **Exposure.** These dates were untouched for this test only. The later gradient and station tests reused them, so those are contrasts on exposed dates (section 21).
- **Independent reproduction.** A separate scorer with its own frame and Earth-orientation code (`validation/oracle`) reproduces all five URE and 3D scores within 0.1 mm (mean ratio 0.766, agreeing to six decimals). An independent review reproduced them as well (section 20).

Evidence: `Opus_Fork/experiments/untouched_dates/` (`PROTOCOL.md`, `RESULTS.md`, `scores.json`); `research/astra_followup_review_20260930/`; `validation/oracle/`.

12. Tropospheric gradients

Why gradients. The linear transfer model predicted about 0.7 cm URE from station gradients of 0.5 mm, an assumed error size (section 10). We therefore tested external, fixed gradient fields:

- **GRAD operational:** TU Wien weather-model gradients, 6-hourly [R6, R12];
- **GPT3:** an empirical climatology [R5].

Test (change against the default)	GRAD operational	GPT3
Aug 3 (development day; used for selection)	URE -4.2%, 3D -2.7%	URE -1.5%, 3D -0.8%
Five validation dates (pre-registered)	URE -1.2, -1.7, -2.1, -0.4, -4.1%: mean -1.9%; 3D mean -2.4%	URE mean -0.61%; 3D mean 0.0%
Reading	**validated on these dates**	inconclusive
Jun 27 (matched default arm added afterwards)	URE -3.8% (2.86 to 2.75 cm; NGA PE 3.79 cm); 3D -2.9%	-

Size and spread. The mean URE gain is 0.59 mm; the smallest daily gain is 0.12 mm (Jul 6). Per satellite, 109 of 154 satellite-days improved in 3D RMS; the worst worsened by 1.98 cm (G24, Jun 6).

Controls:

- the first-iteration trajectories are bitwise equal with and without gradients, since gradients enter only the observations;
- the independent scorer reproduces URE and 3D within 7 micrometres;
- a linear prediction made before any Aug 3 fit matched the refits.

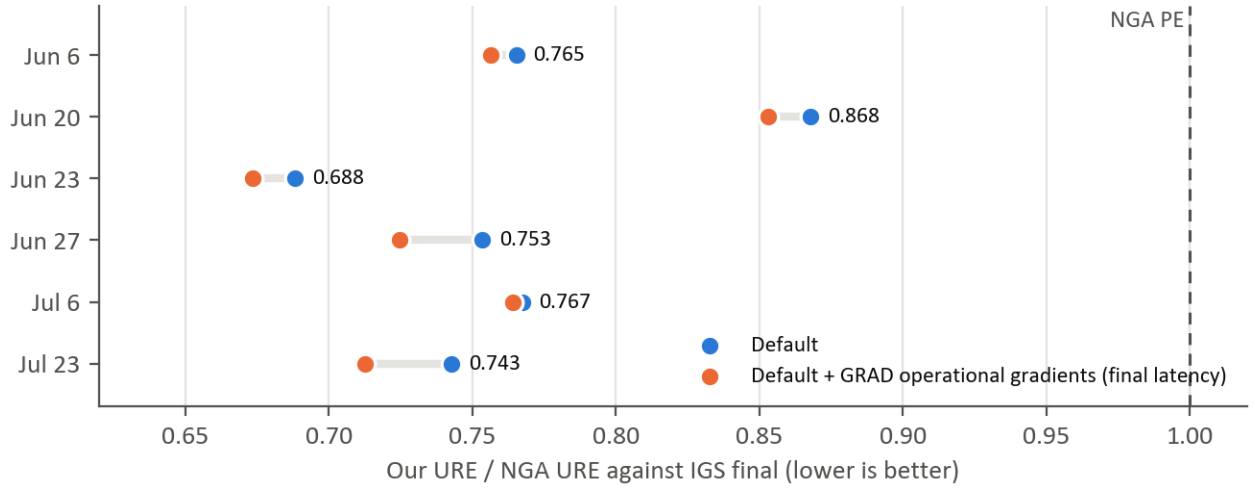
Adoption: optional, labelled, final latency only.

- **Why not rapid.** The last GRAD file an arc needs carried a server Last-Modified time about 40 h after the arc ended. That was read later; it is not an observed first availability or a verified issue time.
- **What the default does.** It stays without gradients, for anything that claims rapid latency.
- **Labelling.** Products made with gradients name the model in the header and use a local "FIN" file label, which certifies no IGS class or delivery time.
- **The rapid route** would be TU Wien's forecast gradients, which need credentials.

About Jun 27. A "sequentially amended comparison": the gradient arm was scored first, and the matched default arm was added afterwards.

Evidence: `Opus_Fork/experiments/gradients/` (`VALIDATION_PROTOCOL.md`, `VALIDATION_RESULTS.md`, `NEW_DATE_RESULTS.md`).

13. Gradients on the six-date set



URE ratio to NGA PE on the five validation dates and Jun 27, for the default (blue) and the default with GRAD operational gradients (orange). Every point is left of the dashed line: closer to IGS final than NGA PE.

Date	Default: URE ratio	+ gradients: URE ratio	Gradient change: URE	Gradient change: 3D RMS
Jun 6	0.765	0.757	-1.2%	-2.0%
Jun 20	0.868	0.853	-1.7%	-1.6%
Jun 23	0.688	0.674	-2.1%	-1.7%
Jun 27	0.753	0.725	-3.8%	-2.9%
Jul 6	0.767	0.764	-0.4%	-1.7%
Jul 23	0.743	0.713	-4.1%	-5.0%

What the gradients change. They lower the radial error on every date, URE by 0.4-4.1% and 3D RMS by 1.6-5.0%. They do not change the overall picture: our URE is 13-33% below NGA PE's on all six dates, with or without them.

What stays the same. Along-track is where NGA PE remains better on several dates, and gradients help it only a little (-0.8% on Jun 27). The largest remaining gap to NGA PE is therefore along-track. Force modelling (eclipse attitude, empirical terms) and observability are candidates for future tests; the atmosphere is not ruled out (section 22).

14. How often do we beat NGA PE?

The URE ratio says how much closer we are on average. The **epoch share** says how often: the 15-minute epochs at which our constellation URE is below NGA PE's, and separately the individual satellite-epochs. Shares are pooled counts; ratios are means of daily ratios. All values are computed with the independent scorer on each comparison's own satellite-epochs; each reproduces the production URE within 0.1 mm.

Development days (Aug 2-5, 7)	Mean ratio: URE / 3D	15-min epochs below NGA PE (URE)	Satellite-epochs below NGA PE (URE)
Baseline	0.942 / 1.088	304/480 (63.3%)	50.8%
+ Earth radiation and thrust	0.876 / 1.082	367/480 (76.5%)	54.7%
+ sub-daily EOP	0.753 / 0.974	442/480 (92.1%)	63.5%
+ elevation weighting	0.727 / 0.923	456/480 (95.0%)	65.0%

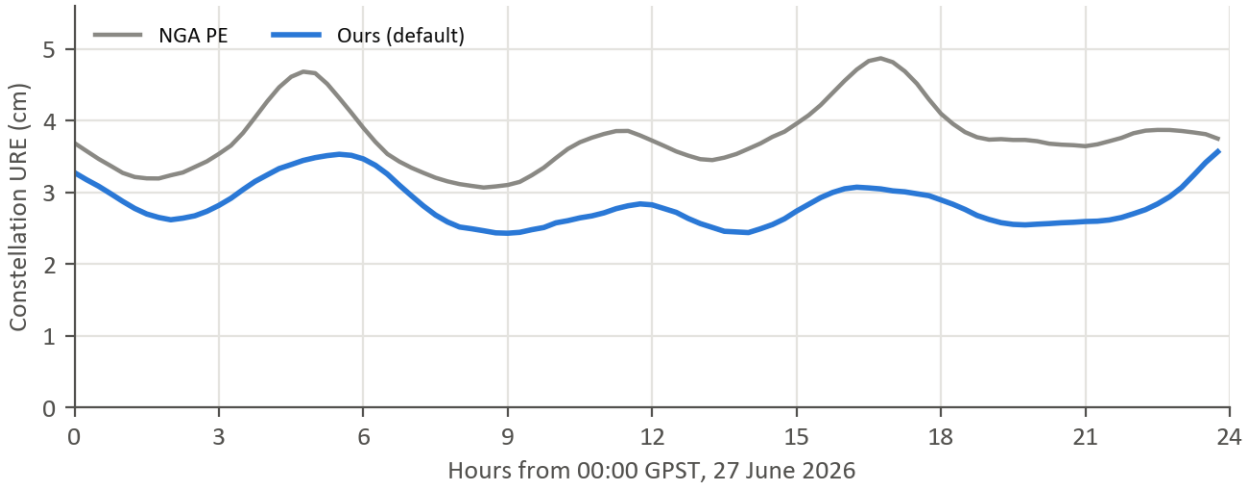
Rejected configurations	Mean ratio: URE / 3D	15-min epochs below NGA PE (URE)	Satellite-epochs below NGA PE (URE)
Free ECOM-7 (Aug 2, 3)	1.138 / 1.220	59/192 (30.7%)	50.1%
ECOM-7 with priors (Aug 2-5, 7)	0.854 / 1.050	395/480 (82.3%)	56.1%
15-degree cutoff (Aug 2-5, 7)	0.745 / 0.958	451/480 (94.0%)	64.4%
Per-station reweighting (Aug 2)	0.982 / 1.204	64/96 (66.7%)	51.8%

What the shares show:

- **The baseline was barely ahead.** It was below NGA PE at 63% of epochs; the elevation-weighted arm at 95%. The sub-daily EOP made the largest single difference in how often we win, from 76% to 92%.
- **Single satellites are close to a coin flip.** The satellite-epoch share stays near 51-65% because, for one satellite at one moment, the two products trade places often. The constellation advantage comes from consistently smaller errors spread over all satellites, not from winning every comparison.
- **The two shares can order differently.** Free ECOM-7 was below NGA PE at only 31% of epochs but at 50% of satellite-epochs.
- **The 15-degree cutoff was not bad against NGA PE.** It was rejected because it did not beat the elevation-weighted arm on held-out days (URE mean +0.06%).

Scope. These are sampled 15-minute epochs on development days, not continuous time and not a probability for other dates. IGS final is the reference.

15. Hour by hour: Jun 27



Constellation URE every 15 minutes on 27 June 2026, ours (default configuration, blue) and NGA PE (gray), both against IGS final.

Date	Default: URE ratio	Default: epochs below NGA PE (URE)	+ gradients: epochs below NGA PE (URE)
Jun 6	0.765	89/96 (92.7%)	90/96 (93.8%)
Jun 20	0.868	84/96 (87.5%)	85/96 (88.5%)
Jun 23	0.688	88/96 (91.7%)	88/96 (91.7%)
Jun 27	0.753	96/96 (100%)	96/96 (100%)
Jul 6	0.767	83/96 (86.5%)	83/96 (86.5%)
Jul 23	0.743	90/96 (93.8%)	92/96 (95.8%)
Six-date set		**530/576 (92.0%)**	**534/576 (92.7%)**

On the six-date set our constellation URE is below NGA PE's at 92.0% of the sampled epochs; 86.5% on the worst date and every epoch on Jun 27. Individual satellite-epochs are below NGA PE 63.0% of the time (63.8% with gradients). An independent review reproduced these counts.

What the figure shows:

- On Jun 27 both products have twice-daily bumps, ours smaller.
- **The arc edge.** Our error rises in the last hour, where the batch fit has data on one side only. That is consistent with weaker constraint at the arc boundary; longer or overlapping arcs are a proposed test, not an established remedy.

Scope. These are sampled constellation epochs on six summer dates, not continuous time, not a per-satellite probability and not a forecast. The epochs are not independent trials.

16. Inputs and rapid latency

The objective asks for inputs that could plausibly be available within about a day. Every run used files downloaded afterwards. A published model existing in advance is different from the exact file versions used here having existed at a cut-off; none of those versions was verified for a cut-off.

Input	Used for	Rapid-suitable?
Station observations, daily 30 s files	Fit	Plausible: daily files normally appear within hours; receipt times not recorded
Broadcast navigation, day and previous day	Fit (starting orbit, satellite selection)	Plausible: the message is real time; the merged daily file's receipt time not recorded
IERS daily EOP (`finals2000A.all`)	Fit, scoring, export	**Not as used.** A later file with final values; a rapid product would use predicted values. Effect not measured
Published models: IERS sub-daily EOP, DE440s, EGM2008, FES2014b	Fit	Yes: analytical models published in advance
Dated files: IGS20 coordinates and ANTEX, IGS satellite metadata	Fit	**Retrospective, unverified.** The coordinate product is cumulative and updated; the metadata file is from September
TU Wien GRAD operational gradients	Optional fit model	**No.** Server Last-Modified about 40 h after the arc (not a verified issue time); final latency
TU Wien GPT3 gradients	Not adopted	Yes (a climatology)
IGS final, NGA PE	Scoring only, never read by the adopted fit	Not applicable

What has not been shown

- **No as-of replay.** We have not established which exact bytes of each input existed by a given cut-off. That replay is the first step towards a rapid claim (section 22).
- **Predicted EOP and other vintages.** Their effect on the orbit has not been measured.
- **Delivery time.** Guarded runs that pause for machine load and resume are not a delivered product within a fixed deadline. Active computing and elapsed delivery time would have to be reported separately.

17. What each result depends on

Three questions are kept apart: what the fit reads (fit inputs), how the model was chosen (model selection), and when the inputs would exist (latency). Every result also shares the retrospective inputs of section 16.

Result	Fit inputs	Model selection and latency	Status
Default	Station data, broadcast navigation, published models; no reference orbit	Chosen on development days using IGS-scored outcomes; the weighting exponent itself came from our own residuals	Adopted; validated on five dates
Default + GRAD gradients (sections 12-13)	As the default, plus GRAD files	Selected on Aug 3 using IGS scores; GRAD files last modified about 40 h after the arc	Optional; final latency
Station tests S17 and S50 (sections 18-19)	As default + gradients	Descriptive contrasts on reused dates	Nothing adopted
Block-mean D0 (6 h arcs)	Calibrated from IGS rapid: reference-derived	-	Superseded; never in the default
ECOM-7 with priors	Prior width from fits to IGS final: reference-informed	-	Rejected
Reference-fitted force tests; IGS censuses (swing, box-wing, sub-daily EOP)	Fitted to IGS positions	-	Diagnostics only; never fit inputs

Rules we follow

- **Fit inputs and model selection are recorded separately.** A model can have independent inputs and still have been chosen by IGS-scored results. That is a selection effect, and we say so.
- **Reference-derived priors are "reference-informed".** They disqualify a reference-independence claim.
- **In the adopted station-fit pipeline, reference products are read only by scoring code;** its fit code refuses to open them. The historical experiments in the table that used reference orbits are labelled as such.

Evidence: `Opus_Fork/results/CORRECTIONS.md` (C02); `docs/consolidation/configurations.md` (legend and section 5); `docs/DATA_SOURCES.md`.

18. Station tests: design

Two network tests were run with the default + gradients configuration (final latency) on the five validation dates, which already had matched 25-station fits with gradients. The protocol was pinned before any station was chosen (`Opus_Fork/experiments/networks/PROTOCOL.md`, amendments A1-A4).

S17: 17 selected IGS stations near NGA's monitor sites. For each publicly described NGA or USAF site, the nearest usable IGS station: MOBS, UNSA, BRUX, BSHM, GLPS, NRC1, WHIT, PIMO, THTG, HRAO, SYDN, NKLG, NLIB, NIST, SGOC, KOKB, MAJU. This is a proxy for the public site list, not NGA's data or estimator.

- Eight are more than 1,000 km from their site: UNSA 1,304, BSHM 1,664, GLPS 1,317, PIMO 2,576, SYDN 2,245, NKLG 2,819, NLIB 1,772 and SGOC 1,779 km.
- The estimator needs BRUX as its clock datum, so (by an amendment made before any fit) BRUX (404 km) replaced the nearest station HERS (141 km) for Hermitage.

S50: the 25-station network doubled. The 25 plus 25 stations added one at a time, each the station farthest from all those already chosen: STJ3, JOG2, TOW2, BSHM, LAUT, KIRU, UNSA, ALGO, IQAL, DARW, SEYG, SCOR, ALBH, POHN, MAL2, DUBO, CEBR, ALIC, QAQ1, JOZE, CKIS, KARR, WHIT, SYDN, NIST. Jun 6 had 49 (no DAV1) and Jul 6 had 49 (no GUAM); the matched 25-station arms had 24 on those dates.

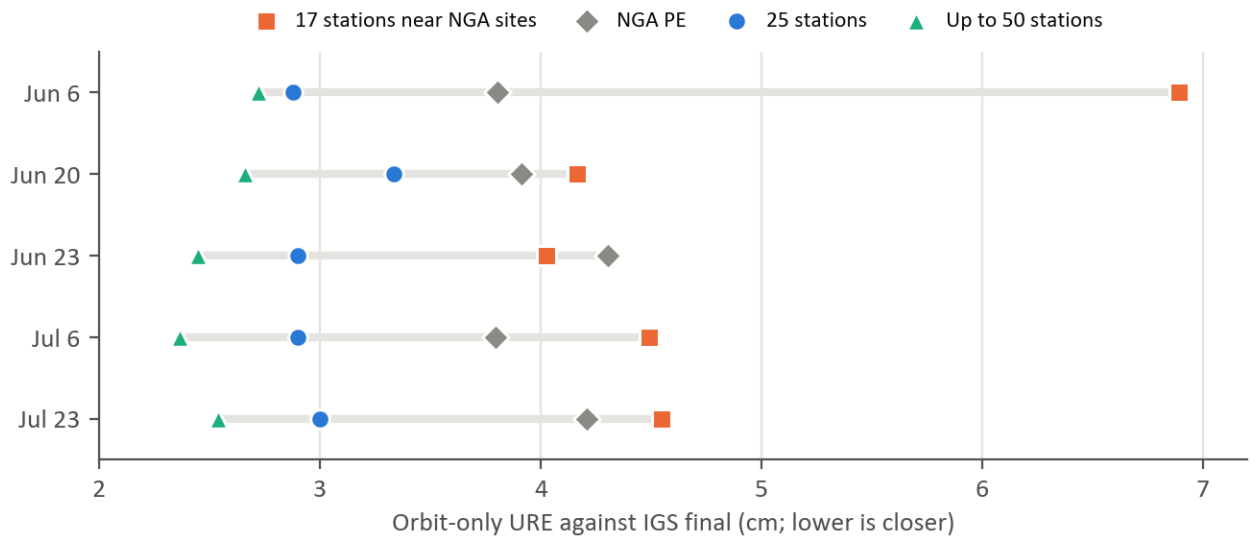
What limits station choice (293 stations listed at BKG on all five dates):

Requirement	Reason
IGS20 coordinate solution valid on the date	123 sites have none; 13 more had expired solutions
No post-seismic deformation model	20 sites excluded (e.g. LPGS, FAIR, DAEJ, WGTN, DGAR)
P(Y)-code signal pair (C1W/C2W)	10 receivers without one (e.g. ASCG) would need code-bias corrections
Supported phase-shift header; exact antenna calibration	9 and 1 rejected
GRAD gradient coverage (optional model only)	4 sites

These are limits of the current preparation code, not of the stations.

A leaner solve for 50 stations. The first S50 fit stopped at its 512 MiB workspace check. Instead of raising the limit, an exact option folds each phase ambiguity out of the square-root system once its arc has ended (a pinned amendment). A real-data parity fit agreed with the standard solver to 0.01 mm (limit 1 mm).

19. Station tests: results



Orbit-only URE against IGS final on each validation date, for 17 stations (S17), NGA PE, our 25 stations and up to 50 stations (S50). All our arms use the default plus GRAD operational gradients.

Date (URE / 3D RMS, cm)	NGA PE	S17	25 stations	S50	S50 vs 25
Jun 6	3.80 / 10.30	6.89 / 21.21	2.88 / 8.99	2.72 / 8.33	-5.4% / -7.4%
Jun 20	3.91 / 10.22	4.17 / 13.00	3.34 / 9.30	2.66 / 7.48	-20.2% / -19.5%
Jun 23	4.31 / 10.63	4.03 / 13.03	2.90 / 8.28	2.45 / 6.79	-15.6% / -17.9%
Jul 6	3.80 / 10.55	4.49 / 14.59	2.90 / 8.90	2.36 / 6.65	-18.5% / -25.3%
Jul 23	4.21 / 10.82	4.55 / 13.87	3.00 / 8.34	2.54 / 6.53	-15.3% / -21.7%
Mean	**4.01 / 10.50**	**4.83 / 15.14**	**3.00 / 8.76**	**2.55 / 7.16**	**-15.0% / -18.4%**

- **S17 does not stay closer to IGS final than NGA PE.** Below NGA PE on 1 of 5 dates in URE and none in 3D RMS; mean ratios 1.215 (URE) and 1.444 (3D); below NGA PE at 40% of 15-minute epochs (25 stations: 91%). Its URE is 62% higher than with 25 stations.
- **S50 "improves"** under the pre-registered rule: better on 5 of 5 dates, mean URE -15.0% and 3D -18.4%, radial lower on every date. Below NGA PE at 99.6% of epochs. Per satellite, the worst got 4.17 cm worse in 3D (G26, Jun 20).
- **What it does not show.** Both tests change geometry and composition as well as count, so they do not say which matters, or why NGA PE does better with a network of similar size.

Checks. All five controls passed: independent rescoring of every arm and of NGA PE, bitwise-equal first trajectories, provenance and model identity, and the exact scored population. One control first failed by 0.015 mm because the two codes used different, equally valid ERFA pole-position methods; the method was matched and the 0.1 mm limit kept (section 20).

Evidence: `Opus_Fork/experiments/networks/RESULTS.md`; `networks_analysis.json`; `Opus_Fork/results/networks_*.json`.

20. Engineering and review

Running on one PC

- **Guards.** Campaign fits, evaluations and report builds run inside resource guards (Windows Job Objects) that cap memory, CPU, threads and time. The multi-process profile holds its GPU telemetry for 10 s (a documented limitation). Small support analyses may run unguarded within the same machine limits.
- **The campaign lease.** Lease-holding controllers and guards take it, so two such campaigns cannot run at once; other controllers are not wired to it.
- **Resumable chains.** Fits resume after shutdowns or machine-load stops; completed workers are reused after hash checks.

Checking our own work

- **Pinned protocols.** Every adoption test had a protocol pinned in its campaign records before it ran. Amendments are appended and say whether outcomes existed.
- **Independent scorer.** `validation/oracle` has its own frame, EOP and sub-daily code; it shares ERFA, inputs and conventions. It offers both ERFA pole-position methods: the series method (default, a cross-method check; 0.2 mm at most) and the angles method used by production, for strict reproduction within 0.1 mm.
- **Tests.** 427 unit and regression tests, including negative controls: 425 run directly and 2 check the guard from inside it, all passing at the 8 October release. Some real-data tests skip when local data are absent.
- **The correction index.** `Opus_Fork/results/CORRECTIONS.md` lists every correction to an earlier claim.

Independent reviews

Two OpenAI Codex agents served as independent reviewers. One (Astra) reviewed each checkpoint, reproduced the untouched-date ratios, the gradient gains and the 92% share, wrote a principal-investigator review before publication, reviewed the station-test controls before any score was read, and reviewed this report for its final edition. The other (Sol) found the antenna-table derivative defect. Reviews are in `research/`.

Defects found and fixed (examples)

- **SP3 export.** A sub-daily-EOP solution was exported with the wrong rotation (about 5 cm); export could also overwrite another product. Both fixed, with tests.
- **Derivative stencil.** A numerical derivative crossed an antenna-calibration boundary (one link of 2,777).
- **Integrity.** Truncated arrays could pass a hash check; a repeated model install could apply weighting twice; the scorer's checker could exit 0 after a failure. All repaired with regression tests.

21. What these results do and do not show

The contribution. A reproducible, resource-bounded, self-written estimator with a carefully recorded development and validation process, and a bounded retrospective performance demonstration. Square-root least squares, ECOM and network processing are established methods; no new estimation method is claimed.

Input role	What it is
Direct observations	Station code and phase (RINEX 3)
Initialization	Broadcast navigation
Fixed external models and calibrations	IGS20 coordinates, igs20.atx, IGS satellite metadata, IERS EOP, FES2014b, DE440s, EGM2008
Fitted constraints	Loose state priors, constrained wet delay, BRUX clock datum
Reference products	IGS final and NGA PE: scoring and model selection; never inputs to the adopted fit (historical reference-informed experiments: section 17)

Relative, not absolute. "About 3 cm" is orbit-only URE against IGS final; the 3D RMS is about 9 cm. Shared geodetic inputs can correlate our errors with the reference. Clocks are not estimated as a product, so this is not a positioning or full signal-in-space result.

Dates	First use	Later use
Development days (Aug 2-5, 7)	Development and model selection	All later work on them: contrasts
Validation dates (Jun 6, 20, 23, Jul 6, 23)	Untouched validation of the default (Jun 23 repaired)	Gradient and station tests
Jun 27	Gradient run	Matched default added after the gradient result

Small effects and spread. Five dates do not give a general success probability, and epochs or satellites are not independent replicates. Aggregate gains coexist with local losses: the gradient model worsened one satellite by 1.98 cm and S50 one by 4.17 cm, in 3D RMS. SP3 files round positions to 1 mm.

22. Conclusions, limitations and future work

Conclusions

- **The retrospective bar is met.** On five validation dates drawn by a pre-registered rule, the frozen default scored 3.06 cm URE against NGA PE's 4.01 cm (mean ratio 0.766) and 8.97 cm 3D RMS against 10.50 cm (0.855), below NGA PE on every date in both metrics (Jun 23 after a documented repair).
- **The margin is radial and cross-track.** Along-track is slightly worse than NGA PE's on average (6.45 against 6.27 cm) and worse on three of the five dates.
- **Options and variants.** GRAD gradients add a small, validated gain at final latency (URE -1.9%, 3D -2.4%); 50 stations a larger, descriptive one (URE -15.0%, 3D -18.4%); 17 stations near NGA's sites do not beat NGA PE.
- **Not shown:** rapid latency, absolute accuracy, general superiority to NGA PE, or other seasons (section 21).

Limitations

- **Retrospective inputs.** No rapid-deadline replay; predicted-EOP effect not measured.
- **Along-track.** NGA PE is still better on several dates. The 12-hour breathing (0.16-0.37 ppb) is unexplained.
- **Physical models.** Nominal yaw through eclipses; fixed radiation and thrust parameters and masses; diagonal weighting; libration omitted. Improvements narrow hypotheses; they do not identify unique causes.
- **Numerical assurance.** Reintegration checks cover synthetic and short cases, not the full adopted model at real boundaries. The Earth-radiation variational derivative omits the radial force-magnitude gradient (a recorded approximation, not yet bounded). The covariance is conditional on fixed coordinates and supplied sigmas.
- **Release integrity.** Evaluation and export check solution hashes and fixture bindings, but a strict check of the full model, fixture and EOP identity is not yet enforced for new products. The multi-process guard can release its admission before killed child processes exit.
- **Reproduction.** No isolated end-to-end rehearsal of the public release; tests needing local data skip.
- **Breadth.** GPS only; float ambiguities; one season; fixed networks; Windows-only guards.

Future work, in order

1. **Close the two release-integrity items** above, each with negative controls.
2. **Rehearse the public release in isolation:** one fit from frozen raw inputs to an exported SP3, outside this PC.
3. **A frozen untouched-date test of a 50-station default** without gradients, if a denser network is wanted.
4. **Rapid as-of replay** with predicted EOP and dated inputs; forecast gradients if credentials allow.
5. **Along-track:** eclipse attitude, empirical terms and longer or overlapping arcs, each under its own frozen protocol.

Why this order. The retrospective bar is met; what limits the claims now is integrity, reproducibility and latency, more than another model term.

23. Evidence index

Topic	Main evidence
Setup, data sources, specification, public release	SETUP.md; docs/DATA_SOURCES.md; docs/REPRODUCING_WITH_AN_LLM.md; docs/KNOWN_LIMITATIONS.md
Every configuration and its status	docs/consolidation/configurations.md
Corrections to earlier claims	Opus_Fork/results/CORRECTIONS.md
Development, 3- and 6-hour arcs	STAGE0_RESULTS.md to STAGE5B_RESULTS.md; YELL_GUAM_RESULTS.md; ARC_LENGTH_RESULTS.md; D0_POD_RESULTS.md; D0_DAY215_RESULTS.md; GUAM_GAP_RESULTS.md
24 h arcs and 25 stations	claude/roadmap/STEP3_RESULTS.md; claude/roadmap/BENCH_RESULTS.md; claude/roadmap/STATIONS_RESULTS.md
NGA PE bar and baseline	Opus_Fork/results/BASELINE.md
Forces and EOP	Opus_Fork/results/T2_1_ERP_RESULTS.md; IIIA_AUG345_RESULTS.md; SDEOP_RESULTS.md; ECOM7_RESULTS.md; ECOM7P_RESULTS.md; ECOM7P_HOLDOUT_RESULTS.md
Weighting and sensitivity	Opus_Fork/results/ELEVATION_RESULTS.md; SENSITIVITY_RESULTS.md; TRANSFER_CALCULATION.md; WEIGHTS_W1.md; WEIGHTS_W2_AND_CAUSE.md
Untouched dates	Opus_Fork/experiments/untouched_dates/ (PROTOCOL.md, RESULTS.md, scores.json)
Atmosphere and gradients	Opus_Fork/experiments/atmosphere/RESULTS.md; Opus_Fork/experiments/gradients/
Time shares and figures	Opus_Fork/experiments/gradients/ure_time_share.json; research/report_rev4/; research/report_rev5/; research/report_rev7/
Station tests	Opus_Fork/experiments/networks/ (PROTOCOL.md with A1-A4, RESULTS.md, networks_analysis.json, scoring_index.json)
URE and 3D tables in this report	research/report_rev7/tables.py and tables.json (read from the recorded evaluations)
Independent reviews	research/*review*/REVIEW.md, notably research/pi_review_20261006/, research/astra_network_review_20261008/ and research/final_report_review_20261016/

Scores quoted in this report come from the recorded evaluations in `Opus_Fork/results/` (`arm_*.json`, `gradval_*.json`, `newdate_*.json`, `networks_*.json`) and `untouched_dates/scores.json`. Paths are relative to the repository root.

24. Sources, terms and document control

References

- **R1.** IGS, Products: rapid and final latency. igs.org/products. Rechecked 6 Oct 2026 for the 17-hour rapid description.
- **R2.** GROOPS cookbook, GNSS network processing; architecture context only. Cookbook. Rechecked 6 Oct 2026.
- **R3.** Petit G, Luzum B (eds) (2010) IERS Conventions (2010). IERS Technical Note 36, Verlag des BKG.
- **R4.** Arnold D et al. (2015) CODE's new solar radiation pressure model for GNSS orbit determination. J Geod 89:775-791. doi:10.1007/s00190-015-0814-4.
- **R5.** Landskron D, Bohm J (2018) VMF3/GPT3: refined discrete and empirical troposphere mapping functions. J Geod 92:349-360. doi:10.1007/s00190-017-1066-2.
- **R6.** Landskron D, Bohm J (2018) Refined discrete and empirical horizontal gradients in VLBI analysis. J Geod 92:1387-1399. doi:10.1007/s00190-018-1127-1.
- **R7.** Steigenberger P, Thoelert S, Montenbruck O (2020) GPS III Vespucci: results of half a year in orbit. Adv Space Res 66:2773-2785. doi:10.1016/j.asr.2020.03.026.
- **R8.** Park RS et al. (2021) The JPL planetary and lunar ephemerides DE440 and DE441. Astron J 161:105. doi:10.3847/1538-3881/abd414.
- **R9.** Pavlis NK et al. (2012) The development and evaluation of the Earth Gravitational Model 2008 (EGM2008). J Geophys Res 117:B04406. doi:10.1029/2011JB008916.
- **R10.** Lyard FH et al. (2021) FES2014 global ocean tide atlas. Ocean Sci 17:615-649. doi:10.5194/os-17-615-2021.
- **R11.** ERFA, derived from IAU SOFA: `xy06` (series) and `xys06a` (angles) CIP methods. liberfa/erfa.
- **R12.** TU Wien VMF data server, GRAD gradients. vmf.geo.tuwien.ac.at.

Full data sources and citation requests: `docs/DATA_SOURCES.md`.

Terms

- **URE:** orbit-only user range error (section 1), the primary metric. **3D RMS:** RMS of the full position error. **RAC:** radial, along-track, cross-track. **EOP:** Earth orientation parameters. **ECOM:** the Empirical CODE Orbit Model. **NGA PE:** the National Geospatial-Intelligence Agency's precise ephemeris.
- **Development days:** Aug 2-5 and 7, 2026. **Validation dates:** Jun 6, 20, 23, Jul 6 and 23, 2026, drawn by a pre-registered rule; their data informed no earlier choice. **Six-date set:** the validation dates and Jun 27.
- **Default:** the frozen configuration of section 4. **Elevation weighting:** phase sigma 9 mm/sin(e)^1.3 (code name elw13). **S17, S50:** the 17- and 50-station test networks (section 18).

Document control

- **Edition.** Final report, Revision 7, 16 October 2026 UTC. It reports results recorded by 8 October 2026; no experiment was rerun for it.
- **What changed from Revision 5.** Presentation only: one metric rule with URE and 3D RMS side by side, consistent names for dates and configurations, section cross-references, scoped claims, and conclusions. The changes follow a combined review by Opus and Astra (`research/final_report_review_20261016/`). New tables and figures are read from recorded evaluations through `research/report_rev7/`. The Jun 27 hour axis was corrected by 10 s in Revision 5 (values unchanged).
- **History.** Revisions 1-5 (PDF) and Revision 6 (an interactive HTML edition, 11 Oct) are archived byte-for-byte in `research/report_history/`. The manifest records the source, evidence and output hashes, and the visual QA of every page.